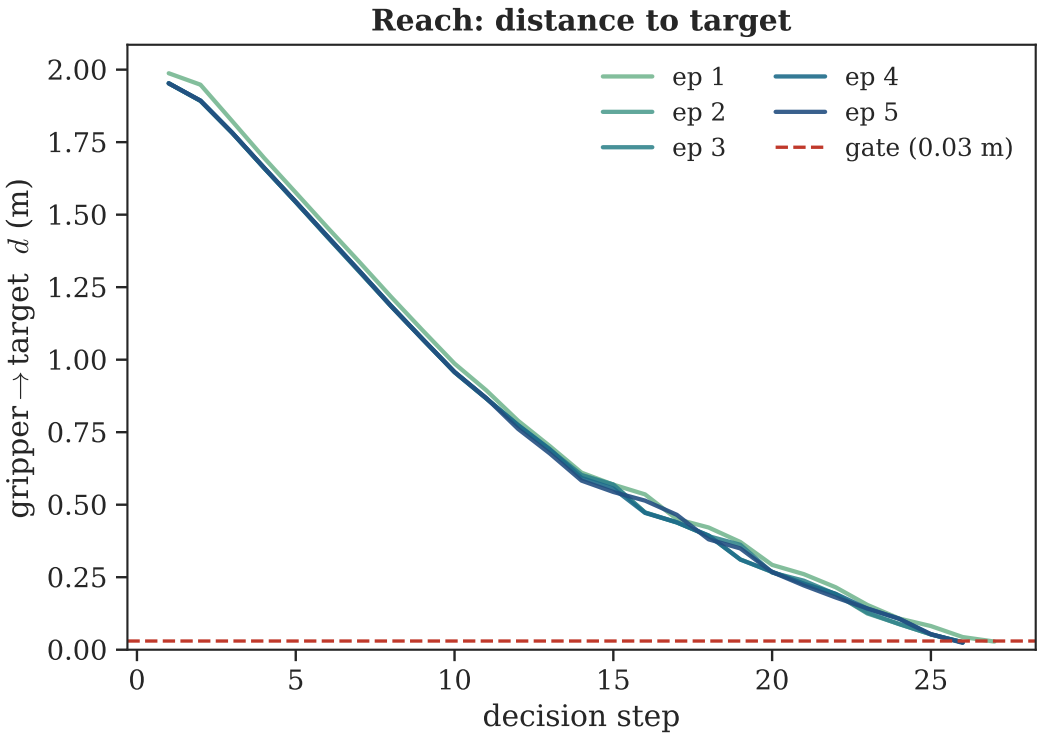
s9_np (fixed target) — Figure Notes for the Defense

5 deterministic episodes, gate 0.03 m / 2.9°, 5/5 success (= 100%); mean min $d = 0.0258$ m, mean min $\varphi = 1.4°$. Date 2026-09-24.

Figure 1 — Reach: distance to target



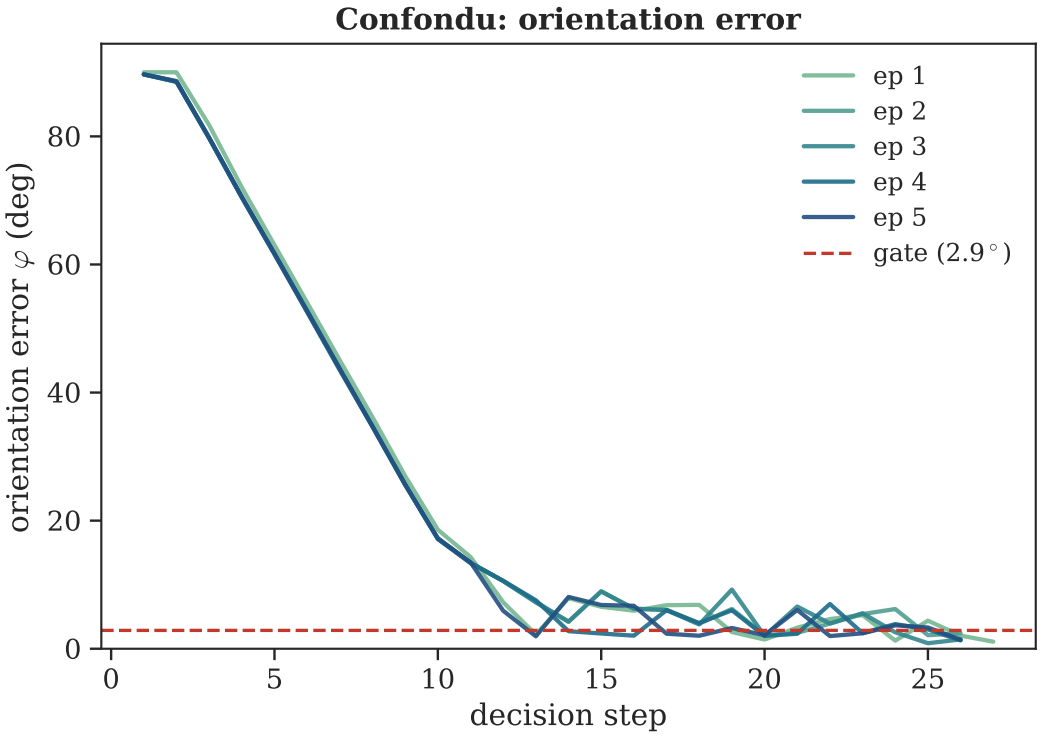
What it shows. Gripper→target distance d vs decision step, one line per episode (5 episodes); the red dashed line is the 0.03 m success gate.

How to read it. All 5 curves fall from ∼2 m to inside the gate in ∼26 decisions and overlap almost exactly. The target is FIXED and the policy deterministic, so the overlap is a reproducibility check.

Say this. "The gripper closes from 2 m to inside the 3 cm gate in about 26 decisions, identically every episode — deterministic and repeatable."

If asked. *Why do the lines overlap?* — Fixed target + `deterministic=True` give the same trajectory each episode; the overlap confirms the run is reproducible.

Figure 2 — Confondu: orientation error



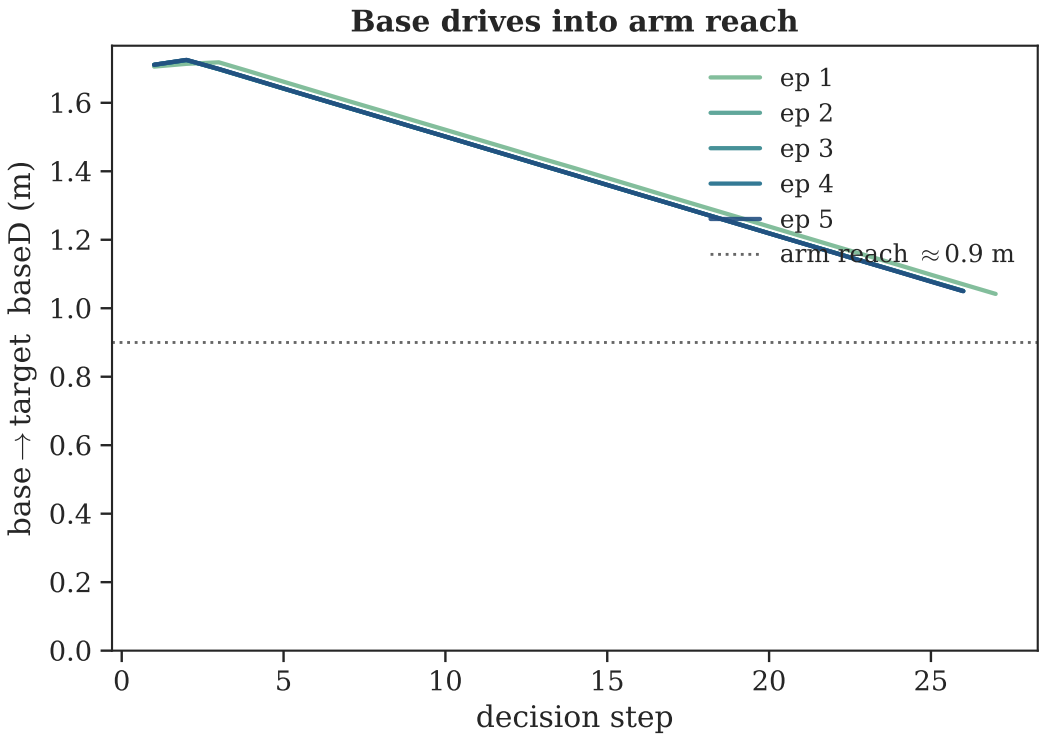
What it shows. Orientation error φ (geodesic angle between the gripper frame and the target frame), in degrees, vs step; gate = 2.9° (0.05 rad).

How to read it. φ starts near 90° (arm home far from the target frame), drops steeply below the gate by ∼step 13, then holds with small residual jitter.

Say this. "Orientation reaches confondu ($< 2.9°$) by ∼step 13 and holds — the wrist aligns well before the reach finishes."

If asked. *What is φ?* — the single geodesic angle between the gripper and target frames (0 = coincident = confondu), from the quaternion dot product $2 \arccos |\langle q_G, q_T \rangle|$.

Figure 3 — Base drives into arm reach



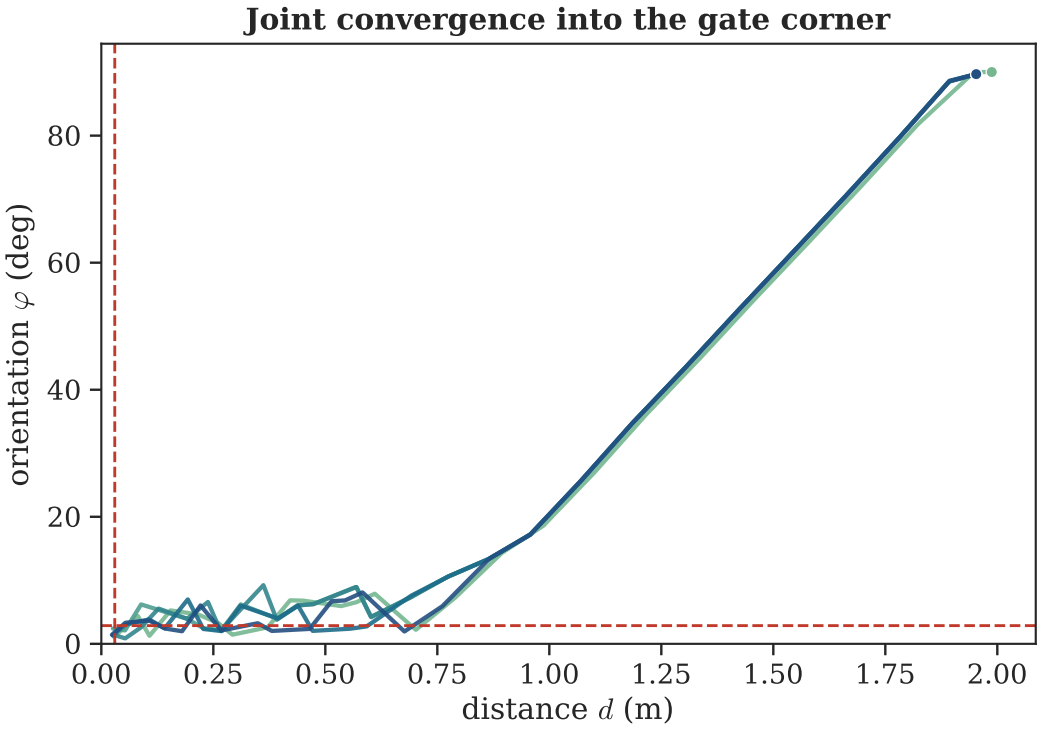
What it shows. base_link→target distance (baseD) vs step; the dotted line marks the arm reach ≈0.9 m.

How to read it. baseD falls from ∼1.7 m to ∼1.05 m — the mobile base drives the target into the arm's working range, then the arm does the fine reach (baseD need not reach 0).

Say this. "The base brings the target from 1.7 m into the arm's range, then hands off to the arm — the whole-body division of labour."

If asked. *Why doesn't baseD reach 0?* — it shouldn't; the base only has to bring the target within arm reach, the gripper (not the base) touches the target.

Figure 4 — Joint convergence into the gate corner



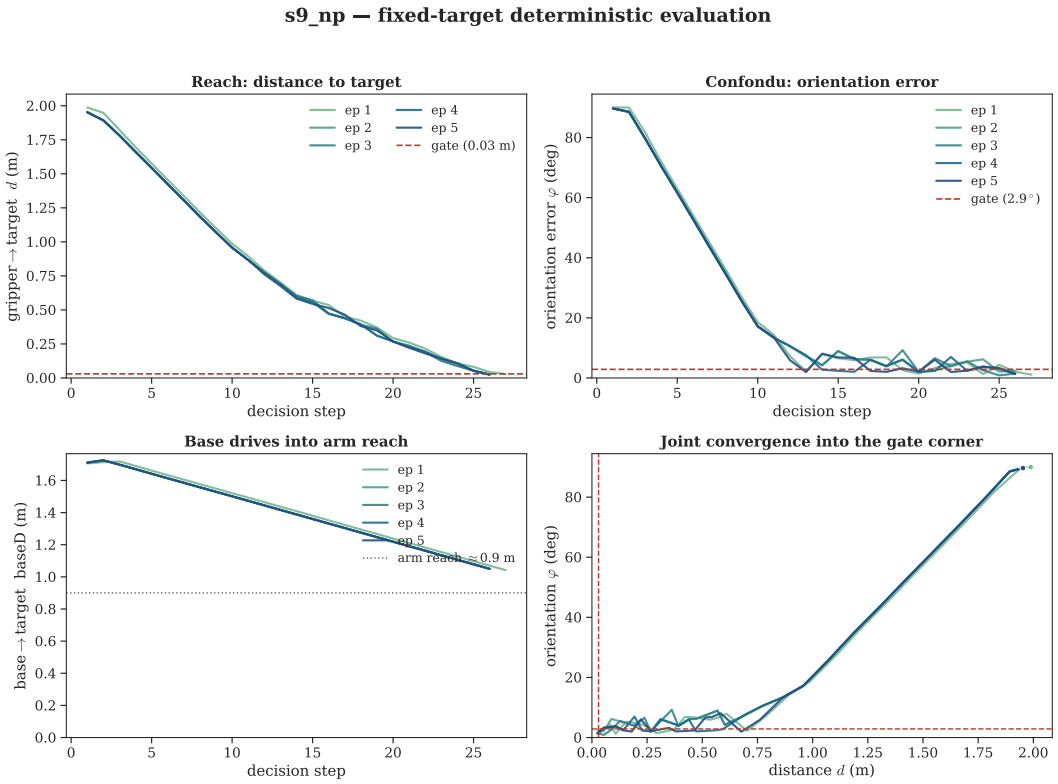
What it shows. The trajectory in (d, φ) space; the dashed lines mark the gate corner ($d < 0.03$ m, $\varphi < 2.9°$).

How to read it. Each run sweeps from the top-right (far, mis-oriented) into the bottom-left gate corner: φ collapses early (by $d \approx 0.8$ m), then d closes the rest at low φ.

Say this. "Both errors converge into the gate corner; orientation is solved first, then position closes — a clean, non-competing convergence."

If asked. *Does solving orientation first cost reach time?* — no; φ is already under gate while $d \approx 0.7$ m, so the wrist just holds alignment while the base finishes driving in.

Figure 5 — Overview (one-slide summary)



What it shows. The four diagnostics on one figure — reach, orientation, base drive, and the (d, φ) phase plot.

How to read it. Use as the summary slide; the panels are detailed on the preceding pages.

Say this. "s9_np reaches the 3 cm gate in ∼26 steps, aligns to $< 2.9°$, drives the base into arm range, and converges both errors into the gate corner — 5/5 episodes, identically."

If asked. *One figure to remember?* — this one: reach, confondu, base drive and joint convergence together.